

CS & IT ENGINEERING



**COMPUTER ORGANIZATION
AND ARCHITECTURE**

Disk

Lecture No.- 1



By- Vishvadeep Gothi sir

Recap of Previous Lecture



doubt?



Topic

Multilevel Cache

Topic

Cache Inclusion Policy

Topics to be Covered



Topic

Magnetic Disk

Topic

Disk Capacity

Topic

Disk Access Time

#Q. Consider a machine with a byte addressable main memory of 2^{16} bytes. Assume that a direct mapped data cache consisting of 32 lines of 64 bytes each is used in the system. A 50×50 two-dimensional array of bytes is stored in the main memory starting from memory location 1100H. Assume that the data cache is initially empty. The complete array is accessed twice. Assume that the contents of the data cache do not change in between the two accesses.

How many data cache misses will occur in total?

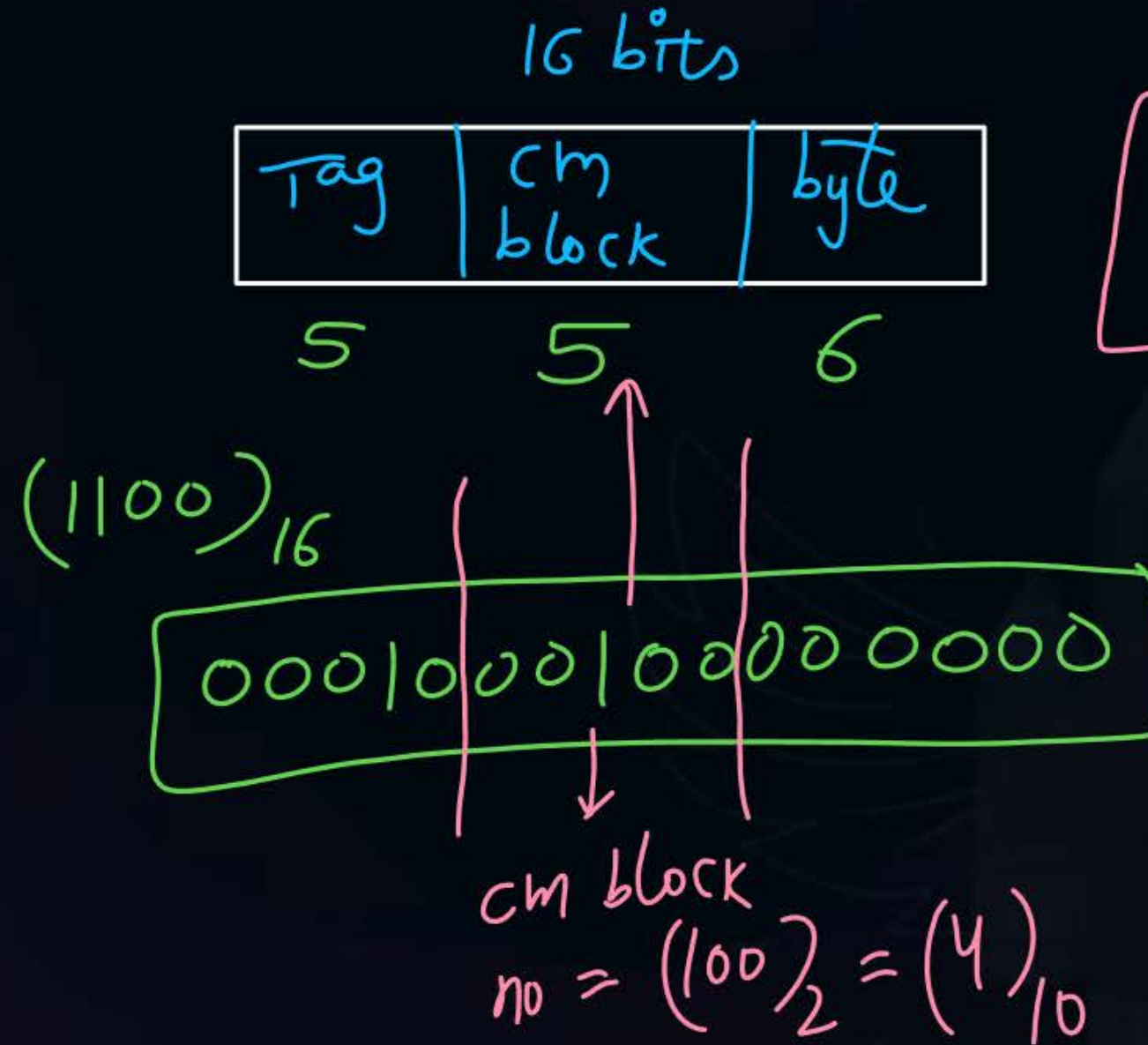
$$\begin{aligned} \text{size of array} &= 2500 \text{ elements} = 2500 \text{ B} \\ \text{blocks to store array} &= \left\lceil \frac{2500 \text{ B}}{64 \text{ B}} \right\rceil = 40 \text{ blocks} \end{aligned}$$

$$\begin{aligned} \text{miss} &= 40 + (40 - 32) \times 2 \\ &= 56 \end{aligned}$$

GATE-PYQ

#Q. Which of the following lines of the data cache will be replaced by new blocks in accessing the array for the second time?

- A** line 4 to line 11
- B** line 4 to line 12
- C** line 0 to line 7
- D** line 0 to line 8



#Q. Consider a direct mapped write back data cache of size 2KB with block size of 128 bytes. The cache is considered to be empty initially. The byte addressable main memory has size 1Mbytes. Further consider that there is an array $A[30][25]$ with each element occupies 4 bytes. The base address of array is $(1A300)_{16}$. The array is accessed 4 times. And between the accesses, there is no any data cache changes happen. Hit ratio of cache for the array access is?

$$\text{size of array} = 30 * 25 = 750 \text{ elements} = 750 * 4 = 3000 \text{ B}$$

$$\text{no. of blocks for array} = \left\lceil \frac{3000 \text{ B}}{128 \text{ B}} \right\rceil = 24$$

$$\text{no. of blocks in cache} = \frac{2 \text{ KB}}{128 \text{ B}} = \frac{2^{11}}{2^7} = 16$$

	1 st access	2 nd	3 rd	4 th
no. of miss	24	$=(2^4 - 16)2$ $= 16$	16	16

$$\text{Total miss} = 72$$

$$\text{Total mem. references} = 750 * 4 = 3000$$

$$\text{no. of hits} = 3000 - 72 = 2928$$

$$\text{hit ratio} = \frac{2928}{3000} = 0.976$$

$$= 97.6\%$$

#Q. A CPU has a 32KB direct mapped cache with 128 byte-block size. Suppose A is two-dimensional array of size 512×512 with elements that occupy 8-bytes each. Consider the following two C code segments, P1 and P2.

P1:

```
for (i = 0 ; i < 512 ; i++)  
{  
    for (j = 0 ; j < 512 ; j++)  
    {  
        x += A[i][j] ;  
    }  
}
```

row wise access

P2:

```
for (i = 0 ; i < 512 ; i++)  
{  
    for (j = 0 ; j < 512 ; j++)  
    {  
        x += A[j][i] ;  
    }  
}
```

column wise access

#Q. P1 and P2 are executed independently with the same initial state, namely, the array A is not in the cache and i, j, x are in registers. Let the number of cache misses experienced by P₁ be M₁ and that for P₂ be M₂.

The value of M₁ is :

$$\text{no. of blocks to store array} = \frac{512 * 512 * 8 \text{ B}}{128 \text{ B}} = 2^{14} = 16384$$

$$M_1 = 16384$$

#Q. P1 and P2 are executed independently with the same initial state, namely, the array A is not in the cache and i, j, x are in registers. Let the number of cache misses experienced by P₁ be M₁ and that for P₂ be M₂.

The value of M₂ is :

$$\text{no. of elements in array} = 2^{18}$$

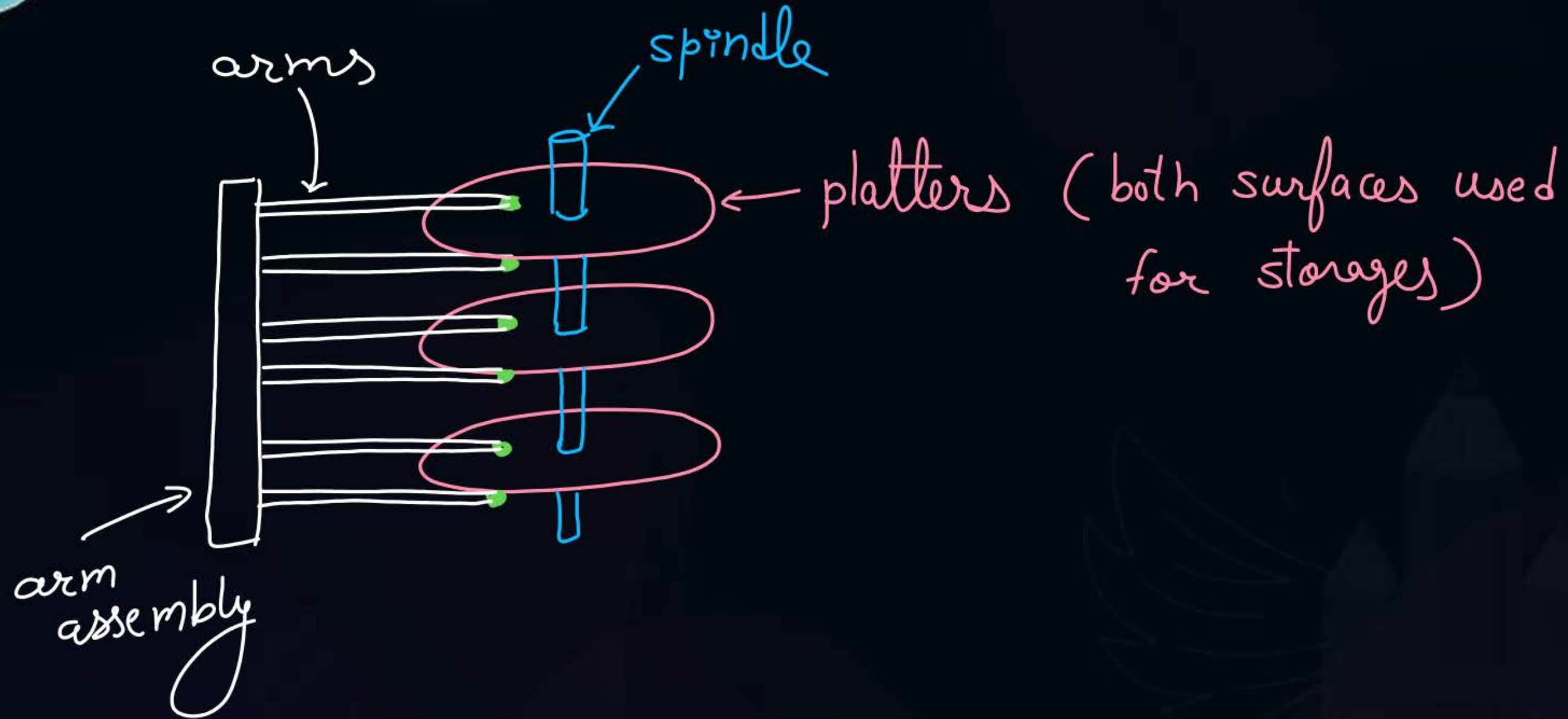
$$M_2 = 2^{18}$$

#Q. Consider a system with 2 KB direct mapped data cache with a block size of 64 bytes. The system has a physical address space of 64 KB and a word length of 16 bits. During the execution of a program, four data words P, Q, R, and S are accessed in that order 10 times (i.e., PQRSPQRS...). Hence, there are 40 accesses to data cache altogether. Assume that the data cache is initially empty and no other data words are accessed by the program. The addresses of the first bytes of P, Q, R, and S are 0xA248, 0xC28A, 0xCA8A, and 0xA262, respectively. For the execution of the above program, which of the following statements is/are TRUE with respect to the data cache?

- A** Every access to S is a hit.
- B** Once P is brought to the cache it is never evicted.
- C** At the end of the execution only R and S reside in the cache.
- D** Every access to R evicts Q from the cache.

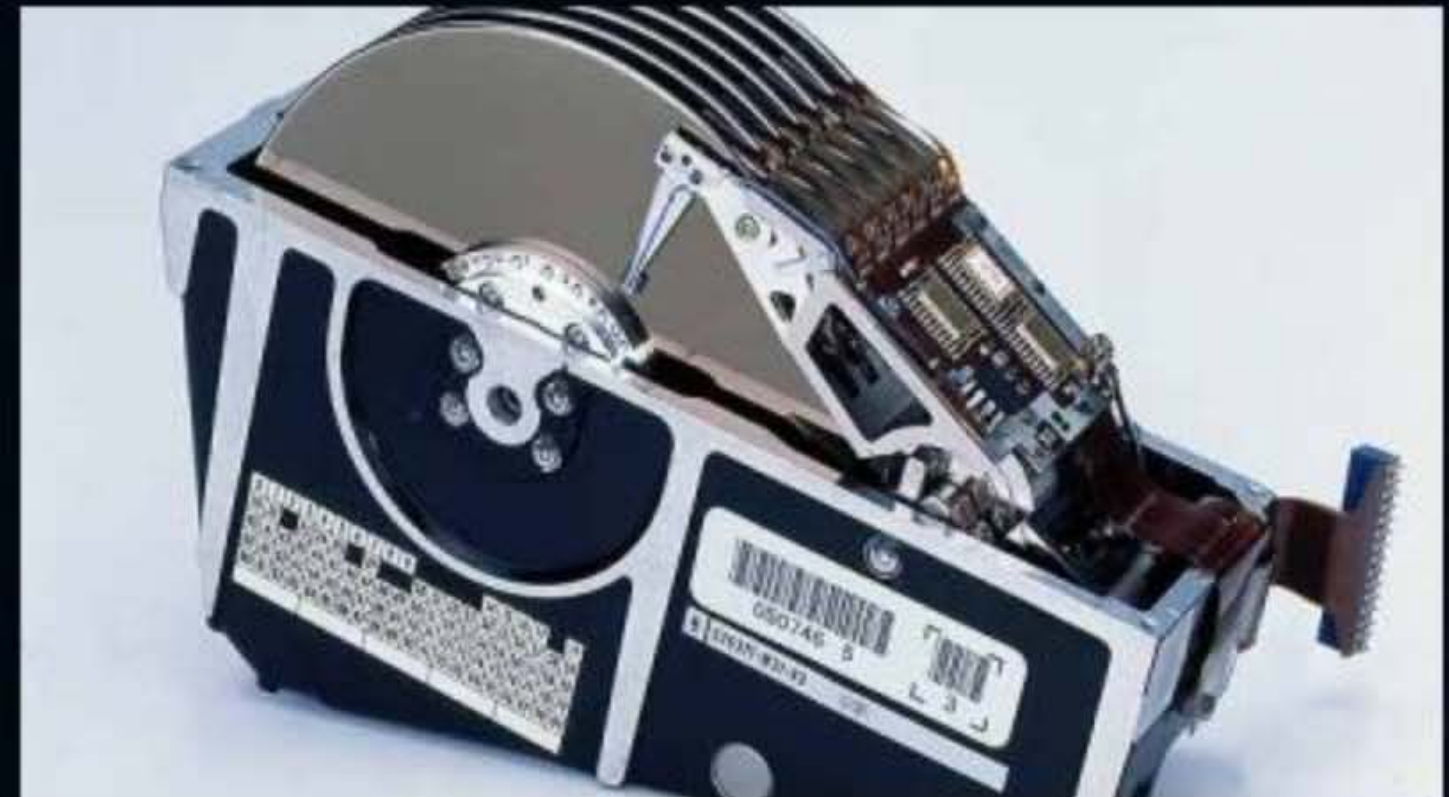


Topic : Magnetic Disk



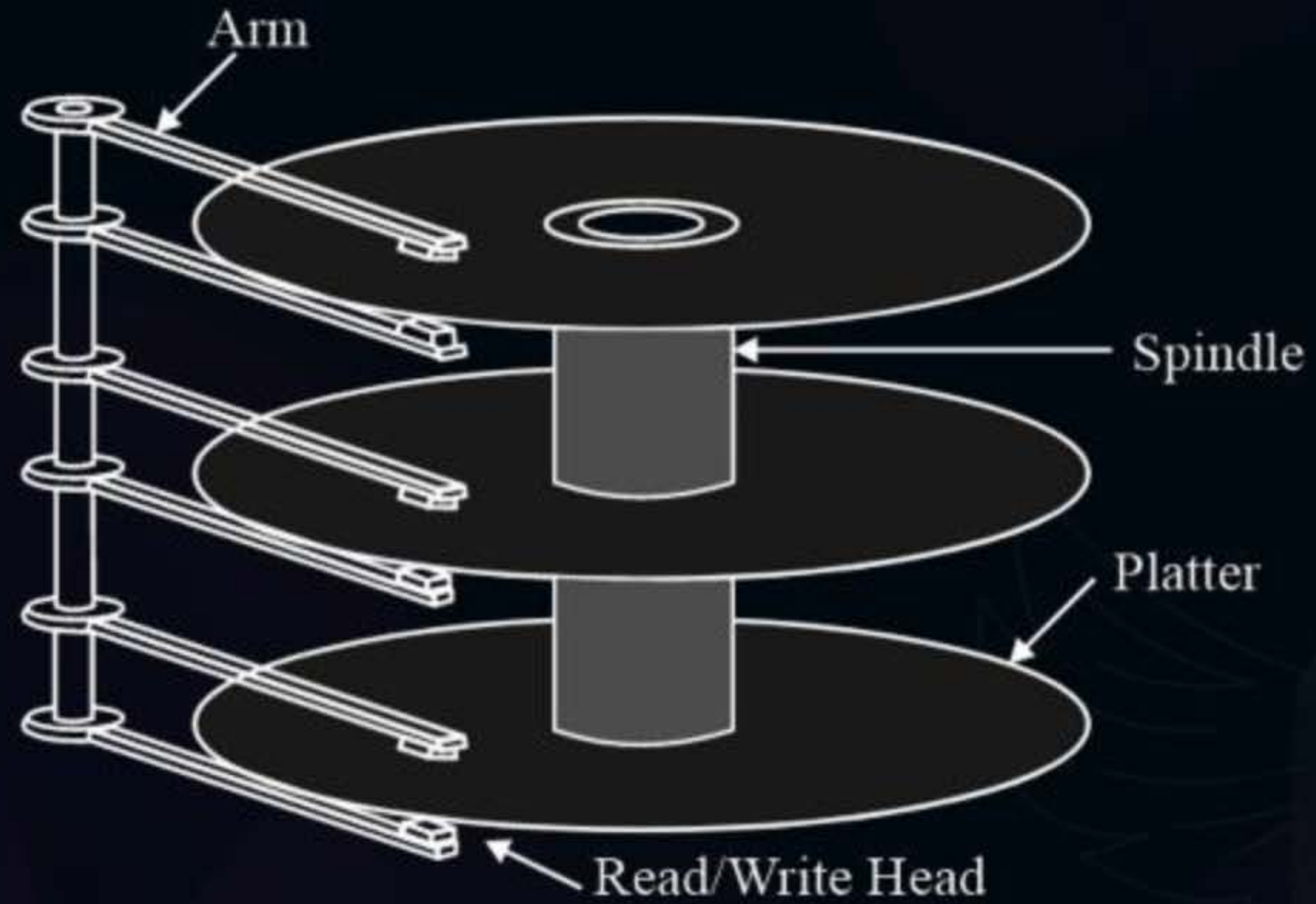


Topic : Magnetic Disk





Topic : Magnetic Disk



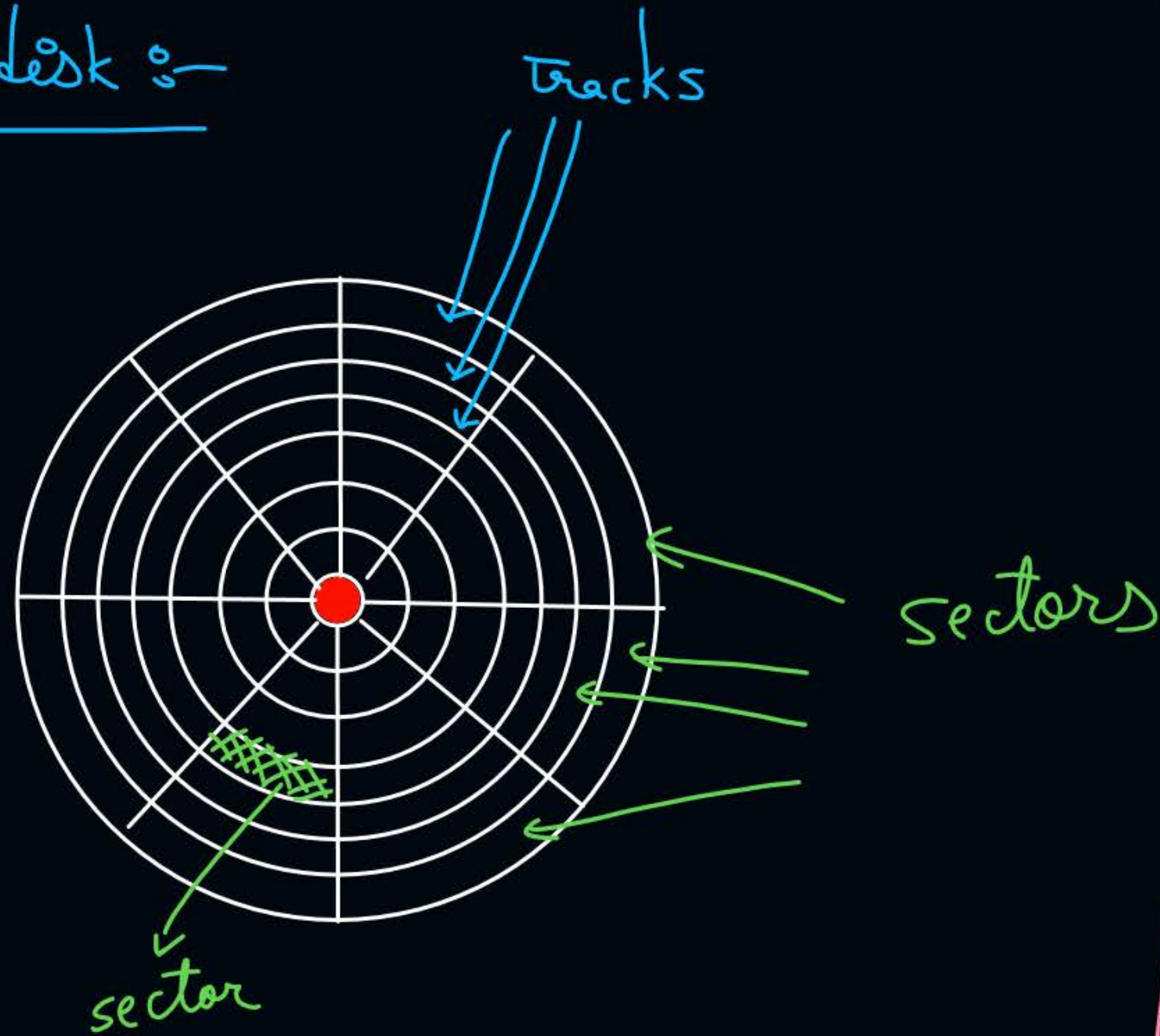


Topic : Magnetic Disk



Number of surfaces in disk:

Top view of disk :-



sector is the smallest part of disk which can be read or write at once.

each sector gets an address in disk



Topic : Magnetic Disk



Number of surfaces in disk: $2 * \text{no. of platters}$

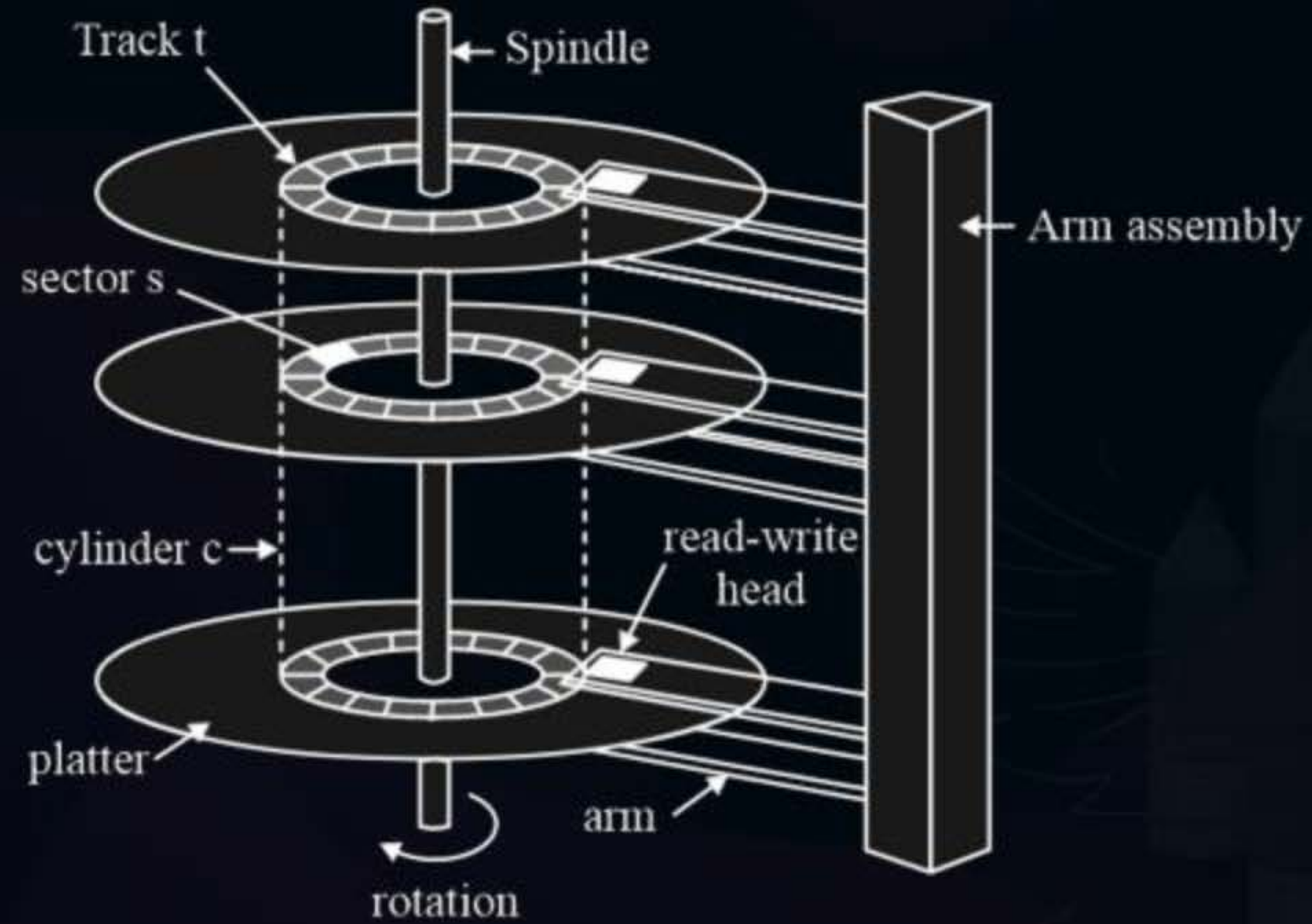
Number of tracks on disk: $\text{no. of surfaces on disk} * \text{no. of tracks per surface}$

Number of sectors in disk: $\text{no. of tracks on disk} * \text{no. of sectors per track}$

Number of bytes on disk: $\text{no. of sectors on disk} * 1 \text{ sector capacity}$
(disk capacity)



Topic : Magnetic Disk





Topic : Sector Capacity

Types of disk

(default)

constant sector capacity
or
variable storage density
or
constant angular velocity

variable sector capacity
or
constant storage density
or
constant linear velocity

#Q. Consider a disk with 32 platters each with 2 recording surfaces. There are 128 tracks per surface and 32 sectors per track. Each sector has equal capacity of 1KBytes.

Calculate:

1. Number of surfaces in disk: $32 * 2 = 64$
2. Number of tracks on disk: $64 * 128 = 2^{13}$
3. Number of sectors in disk: $2^{13} * 32 = 2^{18}$
4. Number of bytes on disk: $2^{18} * 1KB = 2^{28} B = 256 MB$
5. Number of bits for disk addressing: $\log_2(2^{18}) = 18 \text{ bits}$

Disk access time :-

1 sector access time from disk

$$= \text{seek time} + \text{rotational latency} + \text{1 sector transfer time} + \text{additional time}$$



Topic : Disk Access Time

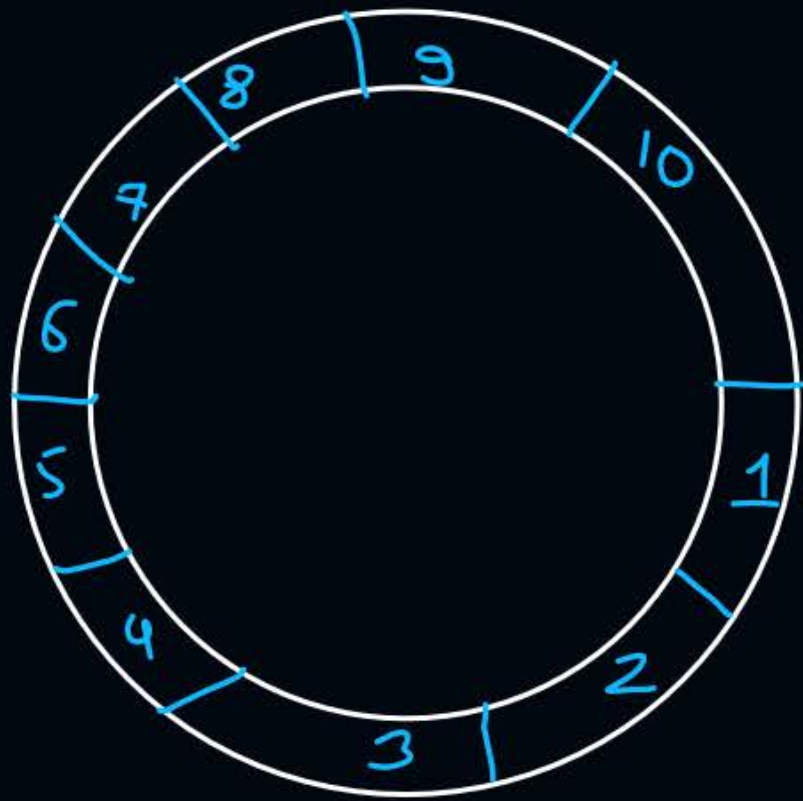
Seek Time: Time required to position the arm over the desired track

Rotational Latency: time required to rotate desired sector under R/W head

Transfer Time: Time required to read or write 1 sector

Note $\Rightarrow$ In one rotation, one track can be transferred.

$$\text{1 sector transfer time} = \frac{\text{1 rotation time}}{\text{no. of sectors per track}}$$



ex.

current position of R/W head	Target is	rotational latency
1 st sector	4 th sector	$3 * 2 \text{ ms}$ $= 6 \text{ ms}$
7 th sector	4 th sector	$7 * 2$ $= 14 \text{ ms}$

no. of sectors per track = 10
 1 rotation time = 20 msec
 1 sector rotation time = $\frac{20 \text{ msec}}{10}$
 $= 2 \text{ ms}$

Avg rotational latency = $\frac{1 \text{ rotation time}}{2}$
 (if current and target
 sector positions not
 given)
 $= \frac{20 \text{ ms}}{2}$
 $= 10 \text{ msec}$

#Q. Disk rotates with 1000 rpm. Find one rotation time?

for 1000 rotations, time = 1 min = 60 sec = 60 * 1000 msec

for 1 rotation, time = $\frac{60 * 1000}{1000}$ ms
= 60 msec

avg. rotational latency = $\frac{60 \text{ msec}}{2}$
= 30 ms

#Q. Consider a disk with 16 platters, 2 surfaces per platter, 2K tracks per surface, 4K sectors per track and 4096 Bytes per sector. Disk rotates with 6000 rpm. Seek time is 5ms. Find disk access time?

1 rotation
time
= 10ms

$$= 5ms + \frac{10ms}{2} + \frac{10ms}{4k}$$

$$= 5 + 5 + 0.0025$$

$$= 10.0025 ms$$

in 0.0025 ms time, disk can transfer data = 4KB

$$\text{in 1 sec time, } \frac{4 \text{ KB}}{0.0025 \times 10^{-3} \text{ sec}} =$$

$$= \frac{4 \text{ MB}}{0.0025 \text{ sec}}$$

$$= 1600 \text{ MB/sec}$$

$$= 1.6 \text{ GB/sec}$$

#Q. A disk has each track with 1k sectors each with 4KB capacity and it takes 10msec for 1 rotation. The transfer rate of the disk is?

$$1 \text{ sector transfer time} = \frac{10 \text{ ms}}{1 \text{ k}} = 0.01 \text{ ms}$$

$$\begin{aligned} \text{in } 0.01 \text{ ms} \quad \text{data transferred} &= 4 \text{ KB} \\ \text{in } 1 \text{ sec} \quad \text{---} \parallel \text{---} &= \frac{4 \text{ KB}}{0.01 * 10^{-3} \text{ sec}} \\ &= 400 \text{ MB/sec} \end{aligned}$$

H.W.

#Q. Consider a disk with 16 platters, 2 surfaces per platter, 2K tracks per surface, 4K sectors per track and 4096 Bytes per sector. Disk rotates with 6000 rpm. Seek time is 5ms. Find the disk transfer rate?





Topic : Where Disk Transfer Rate can be use?



↓
as I/O^s internal transfer rate
to solve I/O organizatⁿ questions





2 mins Summary



Topic

Magnetic Disk

Topic

Disk Capacity

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Disk Access Time



Happy Learning

THANK - YOU

